Quality Management

11 — Interest points: finding the same feature in two images

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Today

Template matching works until the part moves

The simplest way to find a part in an image is **template matching**: slide a stored picture of it over the image and look for the place where they agree best. It fails as soon as the part is **rotated, closer or farther, lit differently or partly covered** — on a production line, that is all the time.

Interest points are distinctive spots that can be found again whatever the size, rotation and light, each with a numerical signature to compare. They are how a machine recognises a part, aligns it to a reference, stitches images and measures in 3D.

1. **What is interesting** — a point you can find again
2. **Blobs at every size** — scale space, Laplacian of Gaussian
3. **The SIFT detector** — extrema, filtering, orientation
4. **Signature and matching** — descriptor, ratio test, RANSAC
5. **In quality control** — align, compare, measure

Parts 1–4 follow S. K. Nayar, *First Principles of Computer Vision* (Columbia University), and D. G. Lowe, “Distinctive image features from scale-invariant keypoints”, *International Journal of Computer Vision* 60(2), 2004. All figures are computed on a synthetic test board with OpenCV.

PART 1

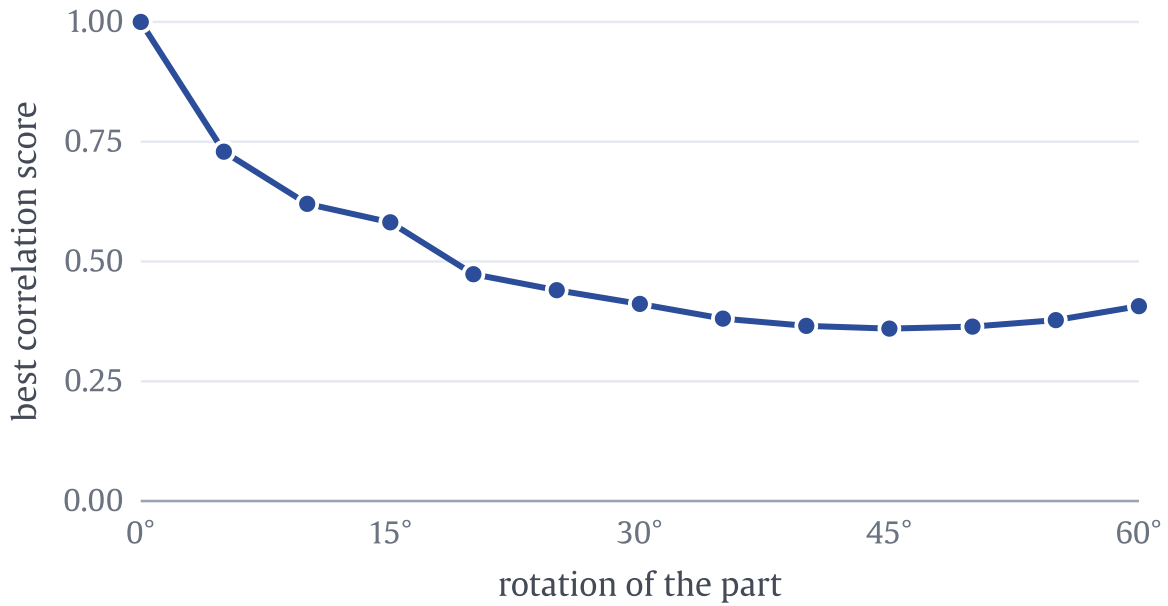
What makes a point interesting

A spot you can find again, whatever happens to the image

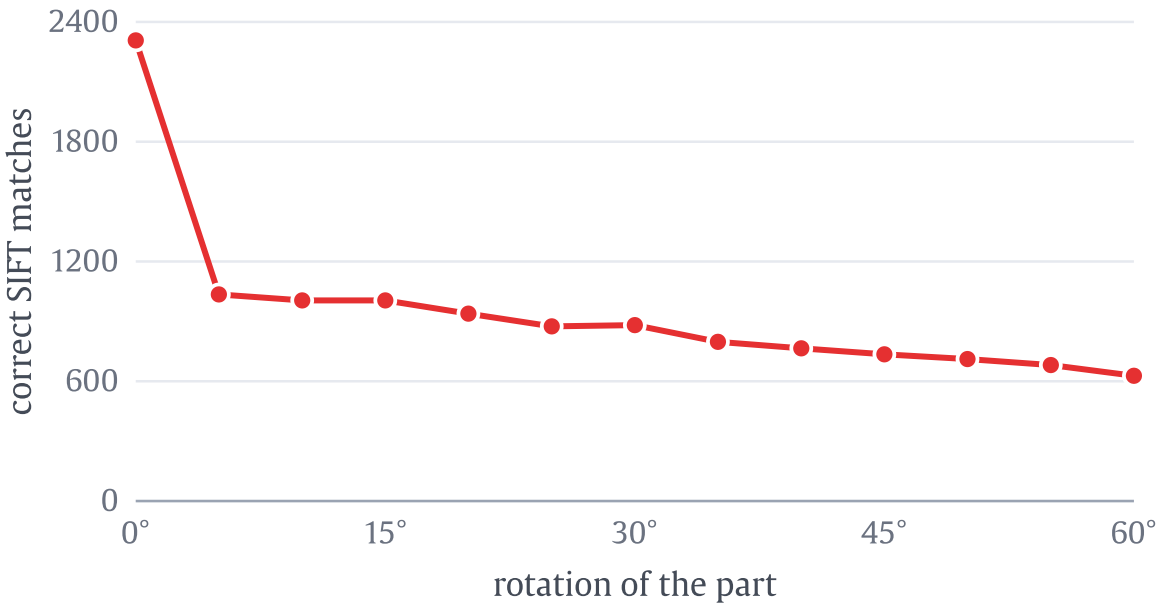
Where template matching breaks

One experiment on our test board: rotate it, and try to find the chip U1 again

Template matching: best agreement with a stored picture of U1



Interest points (this lecture): correct SIFT matches, same images



At 30° the correlation has fallen from 1.00 to **0.41** — no better than a random spot on the board. The interest points still give **881** correct matches at 30° and **627** at 60°.

EVERYDAY · A FACE IN A CROWD

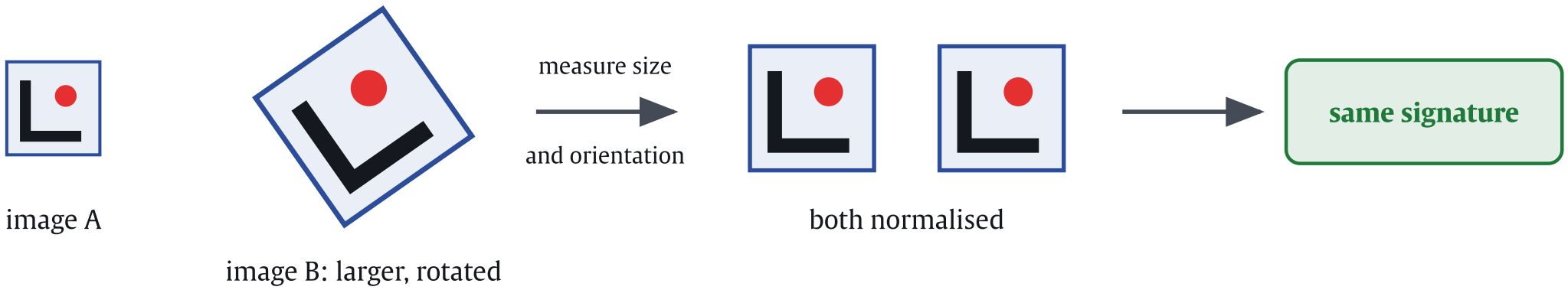
You do not compare a photo with a friend pixel by pixel. You look for a few distinctive details — and you find them whatever the size of the face.

INDUSTRY · PARTS ON A CONVEYOR

A bracket arrives in any orientation and at slightly different distances. A robot must find it and grab it the right way round.

Remove the sources of variation

Measure the size and orientation of a feature, then compare like with like



1. **Find** a point that can be found again, and its precise position.
2. **Measure its size** and cut out a neighbourhood in proportion to it — this removes the scale.
3. **Measure its orientation** and turn the neighbourhood upright — this removes the rotation.
4. **Describe** the neighbourhood with numbers that do not change with brightness — this removes the light.

What makes a point interesting

Five properties a good interest point needs

- **Rich content** — brightness or colour varies inside its neighbourhood. A patch of uniform solder mask has nothing to recognise.
- **A distinctive signature** — a set of numbers that tells it apart from every other point.
- **A well-defined position** — it can be located to within a pixel, or better.
- **Invariant to rotation and scale** — the same point and signature, whatever the size and angle in the image.
- **Insensitive to lighting** — brighter, darker or unevenly lit, the signature stays the same.

Designed on purpose

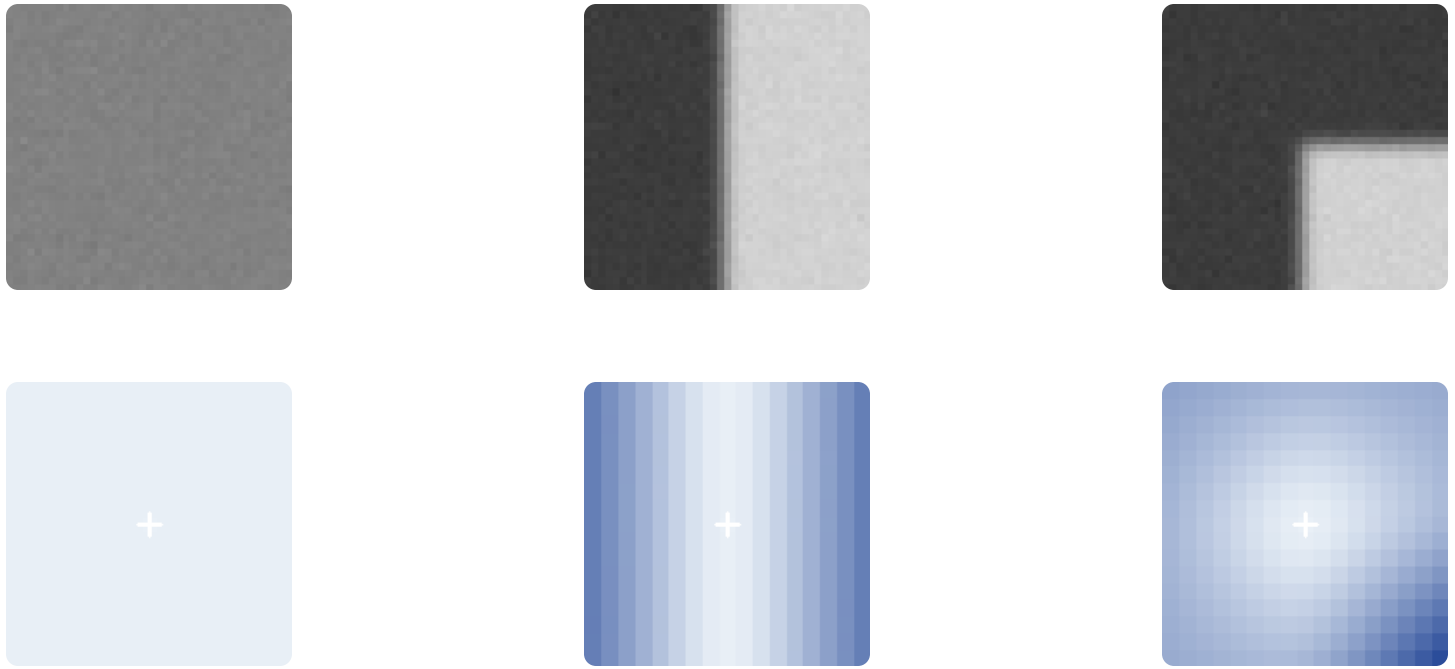
The copper dots in the corners of every printed circuit board — **fiducials** — and the coded targets stuck on parts for photogrammetry are interest points *made* to be found. Natural features have to be found by an algorithm.

DISCUSS

Look around this room. Name a point you could find again from any seat — and one you could not. What is the difference?

Flat, edge, corner

Shift a small window and measure how much the image changes



Top: a 41 × 41 pixel patch. Bottom: how different a 21 × 21 window becomes when shifted by up to 8 pixels in x and y — **darker = more different**; the cross is zero shift.

Flat — nothing changes in any direction: it cannot be located at all.

Edge — changes only across the edge. Along it the window slides freely: the *aperture problem*.

Corner or blob — changes in every direction: one position only. These are interest points.

EVERYDAY

A white wall, a long fence, the corner of a window frame. Which could you point to again, exactly?

INDUSTRY

The straight edge of a sheet-metal part fixes its position across the edge, never along it. A hole or a corner fixes both.

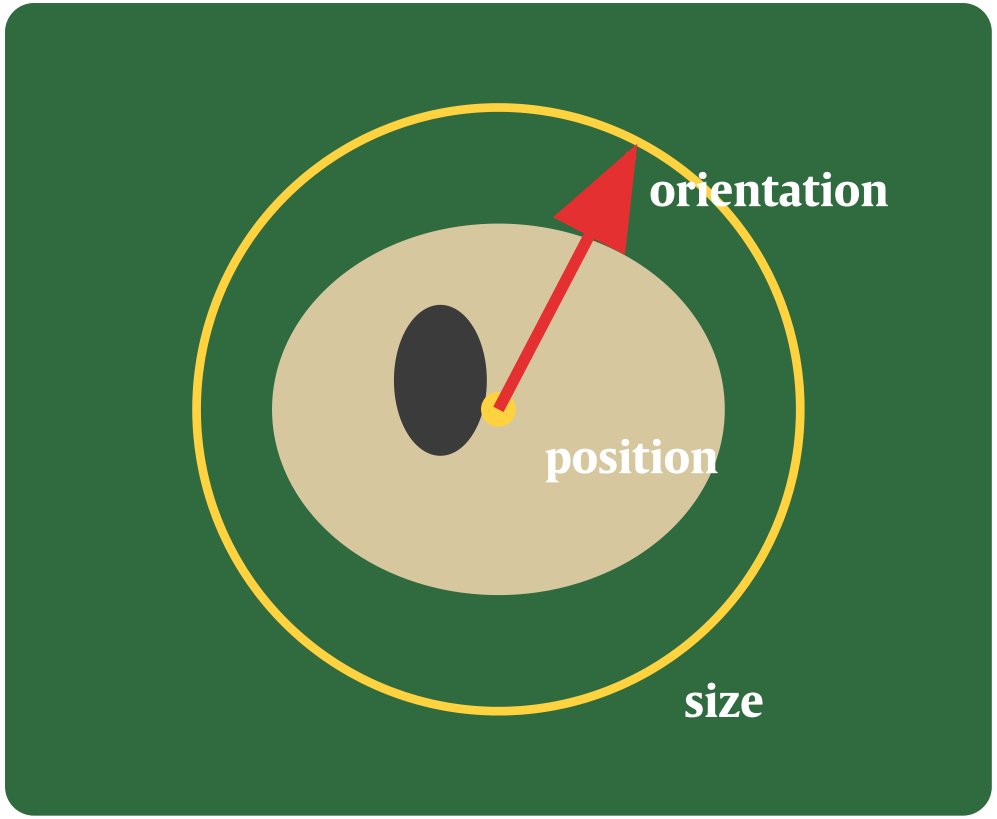
Blobs: a position and a size

The kind of interest point this lecture builds on

A **blob** is a roughly round area brighter or darker than its surroundings: a pad, a via, a hole, a screw head, a printed dot. Unlike an edge, it has a fixed position *and* a definite size.

For a blob to be useful we must:

- **locate** it — its centre;
- **size** it — its characteristic scale;
- find its **orientation** — a direction of reference;
- give it a **signature** that does not depend on size and orientation.



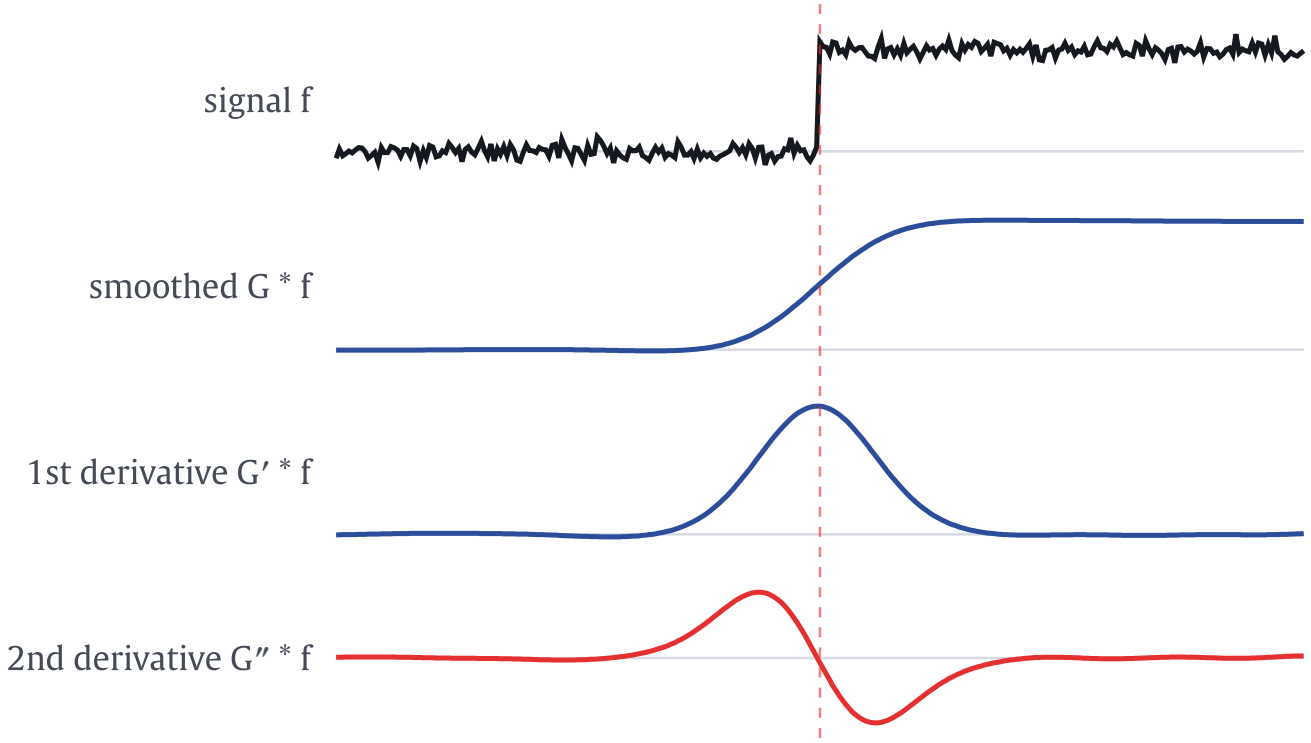
PART 2

Finding blobs at every size

The Laplacian of Gaussian and scale space

Two tools: smoothing and derivatives

Both are filters — the building blocks of everything that follows



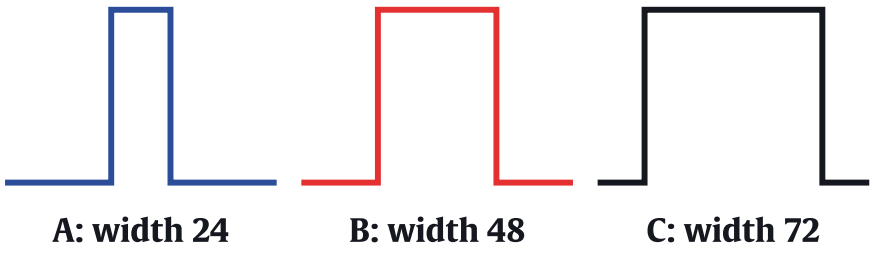
- **A filter** replaces every pixel by a weighted sum of its neighbours; the weights are the *kernel*. Image processing, later in the course, is built on this one operation.
- **Gaussian smoothing** removes noise; its width σ sets how much detail survives.
- **First derivative** of the smoothed signal: a *peak* at the edge.
- **Second derivative**: the signal crosses *zero* at the edge.

The new idea

The second derivative of a Gaussian (G″) also gives a strong dip in the middle of a bright blob — strongest when the blob has the right width for that σ. That is a blob detector.

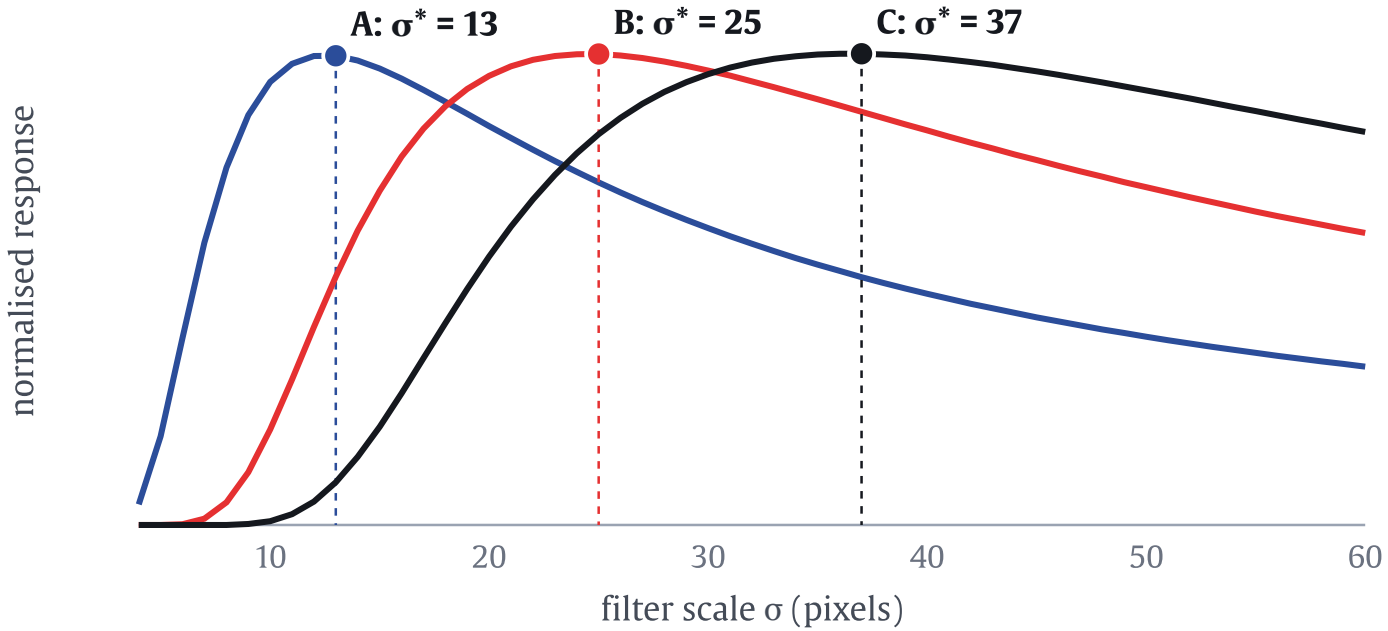
A blob, a filter, and its size

The normalised second derivative answers most strongly when σ matches the blob



Filter each bar with $\sigma^2 \cdot G''_\sigma$ at many values of σ and record the response at its centre.

Without the factor σ^2, wider filters always answer more weakly and the smallest σ always wins. With it, each bar has its own peak: the **characteristic scale σ^***.



Characteristic scale ∝ size of the blob

Widths 24, 48, 72 pixels → $\sigma^* = 13, 25, 37$. Size of A / size of B = σ^*_A / σ^*_B: the filter has measured the size.

In two dimensions: the Laplacian of Gaussian

The 2D blob detector, and the size it finds on a real fiducial

Laplacian: $\nabla^2 = \partial^2/\partial x^2 + \partial^2/\partial y^2$

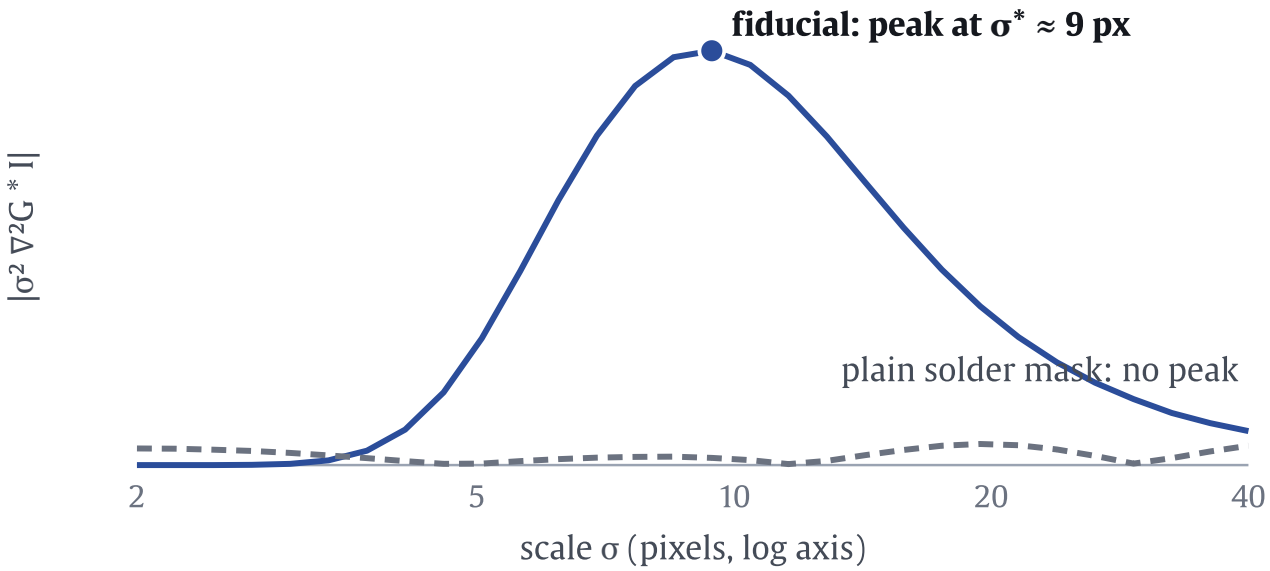
Normalised Laplacian of Gaussian (**NLoG**):

$\sigma^2 \nabla^2 G_\sigma$ — a “Mexican hat” filter.

Convolve the image with it at many σ. A blob is an **extremum in (x, y, σ)**:

$$(x^*, y^*, \sigma^*) = \arg\max \left| \sigma^2 \nabla^2 G_\sigma * I \right|$$

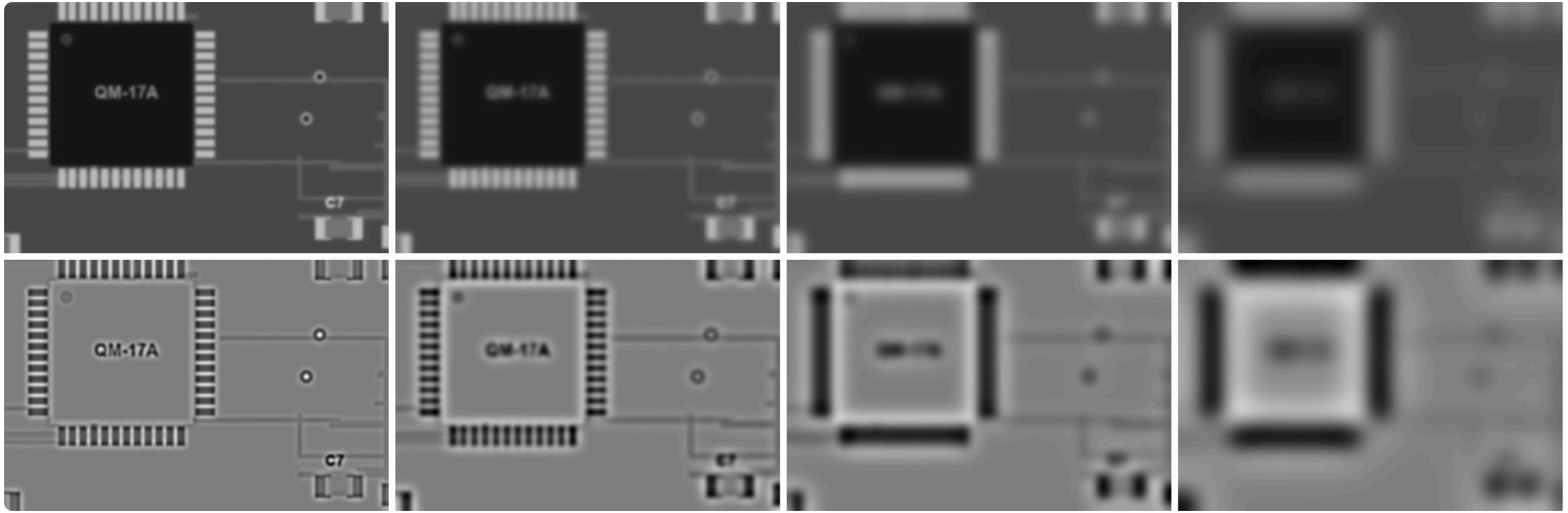
(x^*, y^*) is the position of the blob, σ^* its size. For a round blob of radius r, theory gives $\boldsymbol{\sigma^* = r / \sqrt{2}}$.



On the test board, a fiducial dot of radius 13 pixels peaks at $\sigma^* \approx$ **9.4** pixels; theory says 13 / √2 = 9.2. On plain solder mask the response never peaks: **no strong extremum, no blob**.

Scale space

The same image, blurred more and more — and the differences between levels



Top: $S(x, y, \sigma) = G_\sigma * I$ for σ = 1.6, 3.2, 6.4, 12.8 pixels. Each level is a fixed factor coarser: $\sigma_k = \sigma_0 \, s^k$. Fine detail disappears first — the pins of the chips, then the vias, then the small parts.

Bottom: the difference between neighbouring levels. Grey means no change; bright and dark spots are blobs of the size that level is tuned to. The pins answer at fine scales, the chip bodies only at the coarsest.

Recap TAKEAWAYS

Parts 1 and 2

1. Template matching fails when the part rotates, changes size or light. On the test board its score fell to 0.41 at 30° of rotation.
2. An interest point has rich content, a distinctive signature, a precise position, and is invariant to rotation, scale and light.
3. Flat areas cannot be located; edges only across their direction (aperture problem); corners and blobs in every direction.
4. The normalised Laplacian of Gaussian finds blobs: its extremum over scale gives the position and the size, $\sigma^* \propto$ size.
5. Scale space is the image blurred at a ladder of σ; blobs are the extrema in (x, y, σ).

DISCUSS

A tray of identical washers: every washer is a perfect blob. Why is that bad news, not good news, for matching them between two images?

PART 3

The SIFT detector

Scale-Invariant Feature Transform — D. G. Lowe, 1999 and 2004

SIFT in four steps

A detector and a descriptor in one method

1

Scale-space extrema

difference of Gaussians, compared in 3 × 3 × 3

2

Keep the good ones

drop low contrast and edges; refine below a pixel

3

Orientation

the dominant gradient direction

4

Descriptor

128 numbers per point

Where it comes from

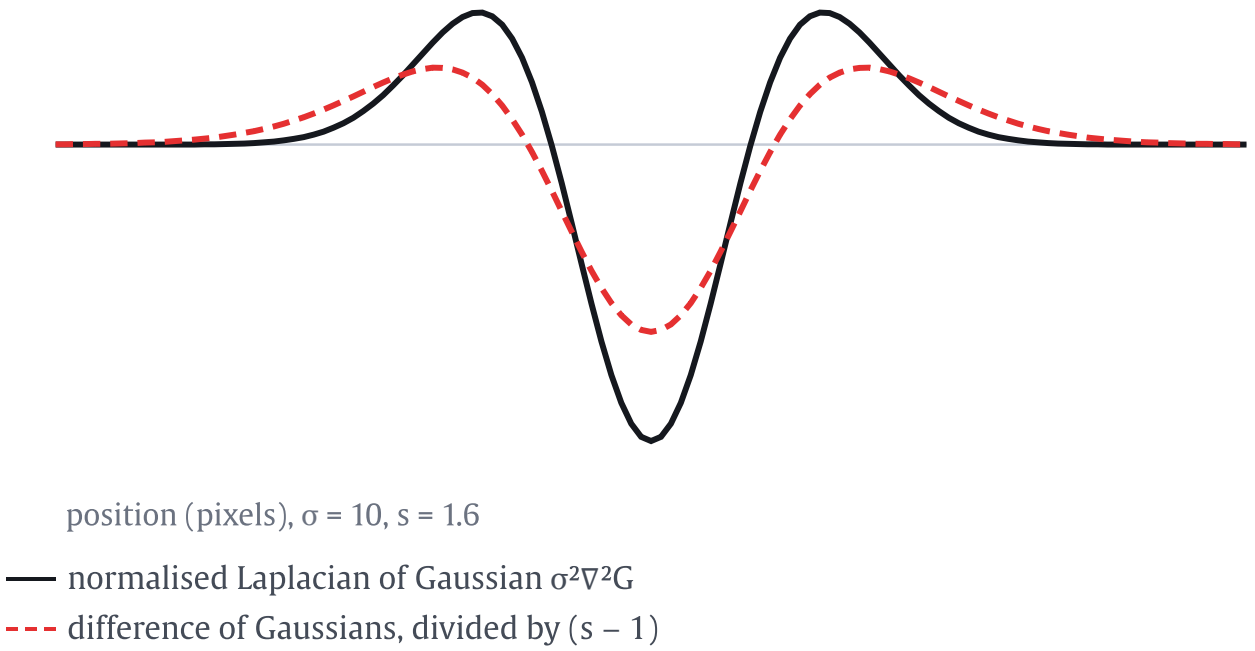
Published by David Lowe (University of British Columbia) in 1999 and, complete, in 2004. One of the most cited papers in computer vision; still the reference against which new detectors are measured.

Free to use since 2020

SIFT was patented until 2020. It is now part of the main OpenCV library, the open-source toolkit used in most industrial and research vision software — and in every figure of this lecture.

A fast Laplacian: the difference of Gaussians

Two blurred images, subtracted, are almost exactly the normalised Laplacian



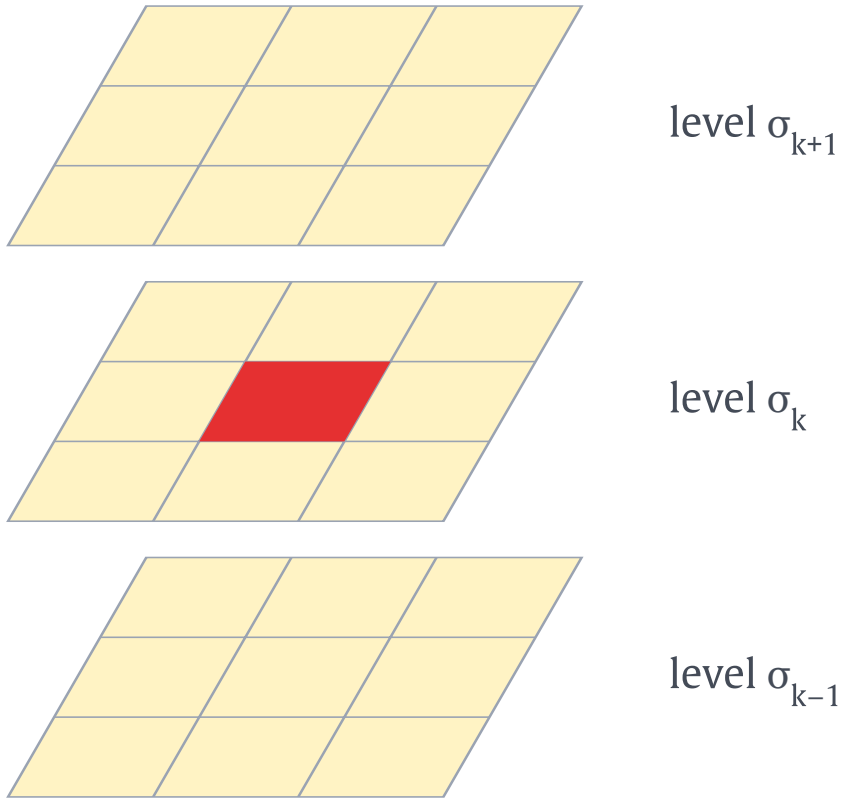
$$\mathrm{DoG} = G_{s\sigma} - G_{\sigma} \approx (s-1)\cdot\sigma^2\,\nabla^2 G_{\sigma}$$

The blurred images of the scale space are computed anyway. Subtracting neighbouring levels costs almost nothing — and gives the blob detector directly.

Lowe's settings: 3 levels per octave ($s = 2^{1/3}$), starting at $\sigma_0 = 1.6$; after every octave the image is halved, so the search over large scales stays cheap.

Finding the extrema

Compare every pixel with its 26 neighbours in position and scale



A pixel of a difference-of-Gaussian level is a **candidate** if it is larger — or smaller — than all its neighbours: 8 in its own level and 9 + 9 in the levels above and below.

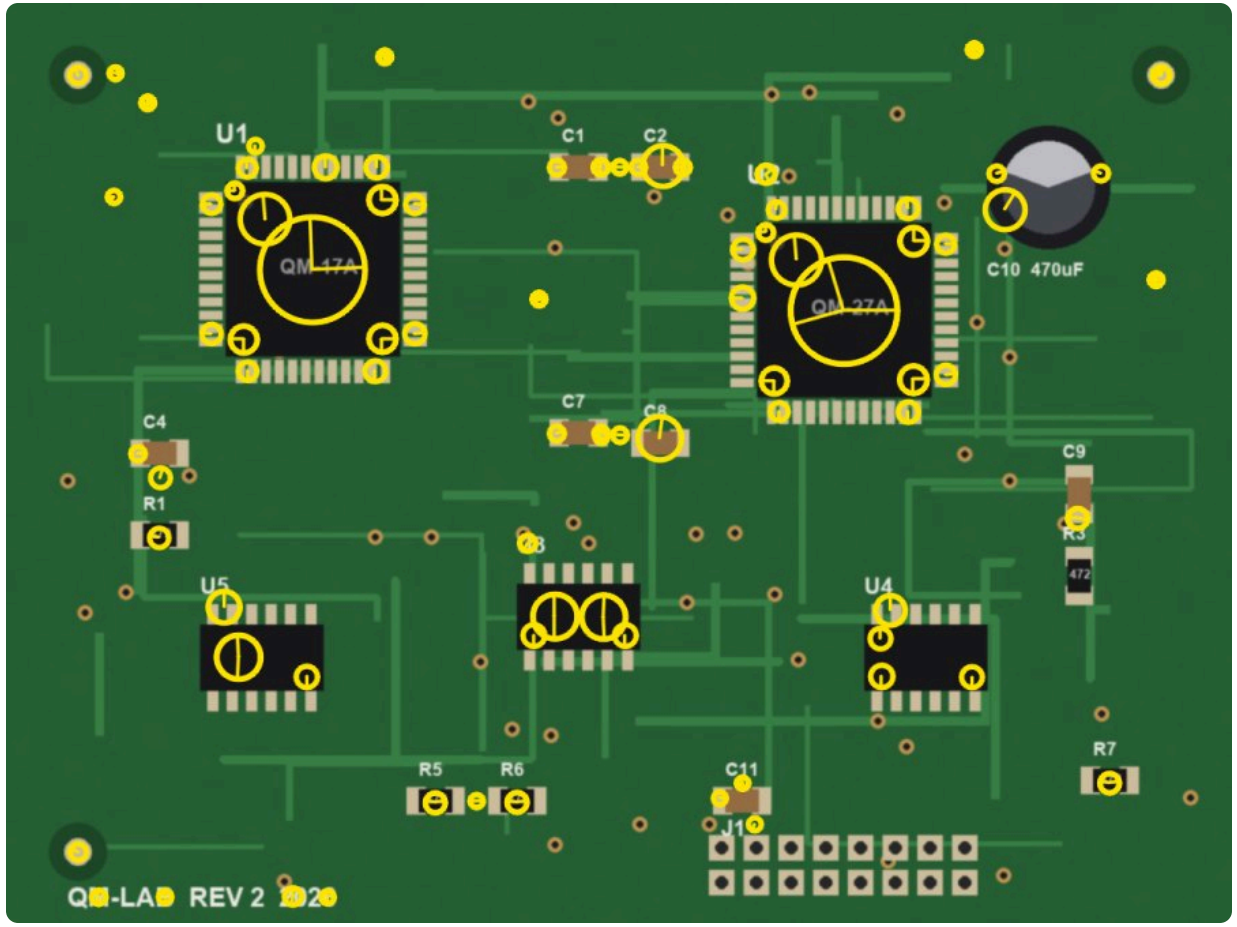
Its position and scale are then **refined between pixels** by fitting a quadratic surface through the neighbours. The interest point is a measurement: located to a fraction of a pixel, which is what makes it useful for alignment and 3D measurement later.

ON THE TEST BOARD

Before any filtering, the search finds **7,289** candidates on a 1200 × 900 pixel image.

Keeping only the good points

Two filters throw away most candidates



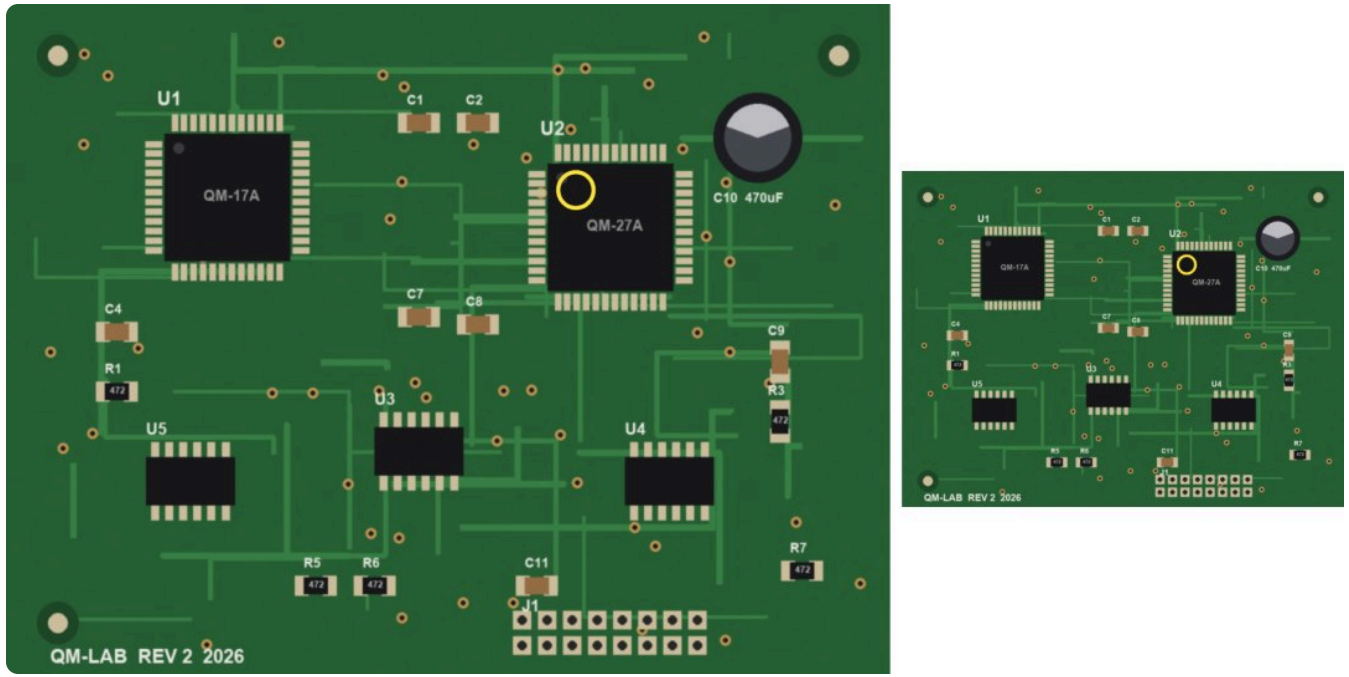
The 150 strongest keypoints big enough to see here. Circle = scale, line = orientation.

Stage	points
Extrema in scale space	7,289
– low contrast	3,315
– edge-like	2,412

- **Low contrast** — a weak extremum (Lowe: $|D| < 0.03$ of the full grey range) is mostly noise: it would not be found again.
- **Edge-like** — strongly curved across, flat along (ratio of principal curvatures above 10). Its position slides along the edge: the **aperture problem** of slide 7, again.

Scale invariance

The ratio of the characteristic scales is the ratio of the sizes



The same board at full size and at half size. One keypoint is drawn in both: its circle is 49 pixels across on the left, 24 on the right. Over all **673** matched points, the median ratio of scales is **2.00** — the true ratio is 2.

EVERYDAY

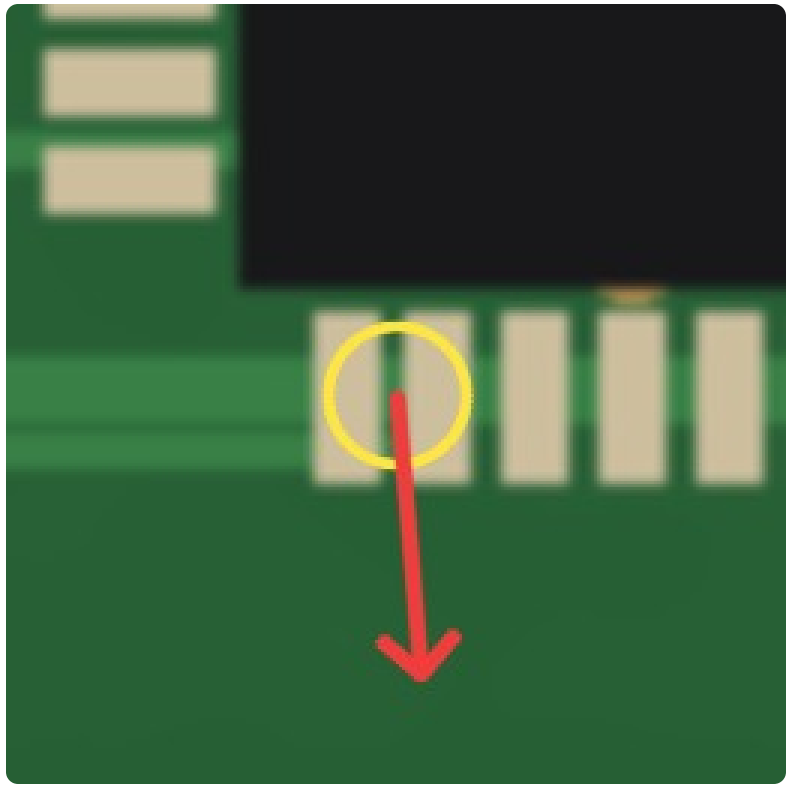
A coin photographed from 30 cm and from 60 cm: every detail of it is found again, at half the scale.

INDUSTRY

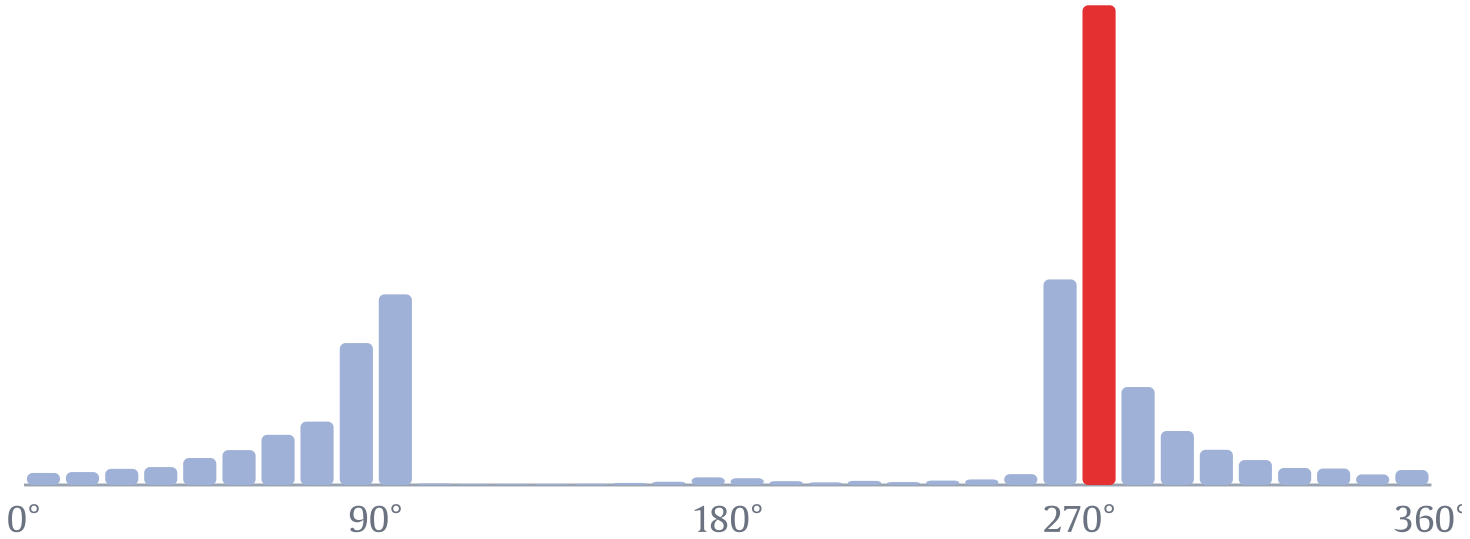
Two cameras with different lenses see the same part: the scale ratio of matched points gives their relative magnification, without a ruler.

Orientation

Turn every neighbourhood to its own dominant direction



One keypoint of the board: circle = scale, arrow = principal orientation.



gradient direction (36 bins of 10°), weighted by strength

- Around the point, every pixel votes for its **gradient direction**, weighted by the strength of the gradient and by its distance from the centre: 36 bins of 10°.
- The highest bin is the **principal orientation** — here 275° (OpenCV, interpolating between bins: 272°).
- Any other bin above 80 % of the highest (dark blue) creates a second keypoint at the same place, with that orientation.

Recap TAKEAWAYS

Part 3 — the SIFT detector

1. SIFT approximates the normalised Laplacian with a difference of Gaussians: two blurred images, subtracted.
2. Candidates are extrema among 26 neighbours in position and scale, refined to a fraction of a pixel.
3. Low-contrast and edge-like candidates are discarded: on the test board, 7,289 candidates became 2,412 keypoints.
4. Each keypoint carries a position, a scale and an orientation. Scale ratios between images are size ratios (measured: 2.00 for 2).
5. The orientation is the peak of a histogram of gradient directions; it makes the description rotation-invariant.

DISCUSS

An edge detector looks for edges on purpose. Why does SIFT throw them away?

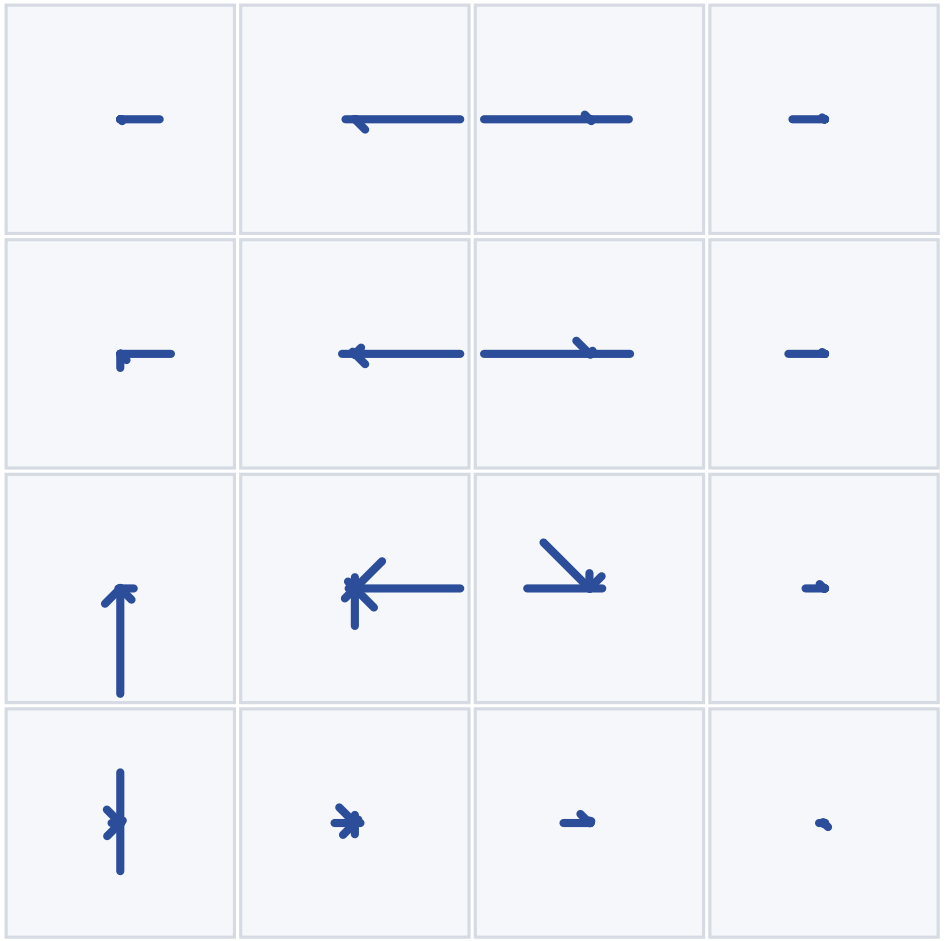
PART 4

A signature, and how to match it

The descriptor, the nearest neighbour, the ratio test and RANSAC

The SIFT descriptor

128 numbers that describe the neighbourhood of a point



The real descriptor of the keypoint on the previous slides. Each cell: the length of a spoke is the weight of that gradient direction.

1. Take the neighbourhood **scaled to σ*** and **turned to the principal orientation**: scale and rotation are gone.
2. Divide it into **4 × 4 cells**.
3. In each cell, build a histogram of gradient directions with **8 bins**.
4. 4 × 4 × 8 = **128 numbers**. Normalise them to length 1, clip the largest, normalise again: brightness and contrast are gone.

Why histograms of directions

They keep the *shape* of the neighbourhood — where the edges are and which way they run — but not exact pixel positions. A small shift, a little blur or a slight perspective change hardly alters them.

Comparing two signatures

Three ways to say how alike two descriptors H_1, H_2 of length N are

L2 distance

$$d = \sqrt{\Sigma_k (H_1(k) - H_2(k))^2}$$

Smaller is better; 0 is a perfect match.
The one SIFT uses.

Normalised correlation

$$\Sigma (H_1 - \bar{H}_1)(H_2 - \bar{H}_2) \,/\, (\|H_1 - \bar{H}_1\| \, \|H_2 - \bar{H}_2\|)$$

Larger is better; 1 is a perfect match. The same measure as template matching, applied to signatures instead of pixels.

Histogram intersection

$$d = \Sigma_k \min(H_1(k), H_2(k))$$

Larger is better: how much the two histograms overlap.

For every keypoint of image A, the matcher finds the descriptor of image B at the smallest distance: its **nearest neighbour**. On the test board: 2,412 keypoints in the reference, so 2,412 nearest neighbours — and most of them are wrong.

A nearest neighbour is not yet a match

Two checks turn candidate pairs into reliable correspondences

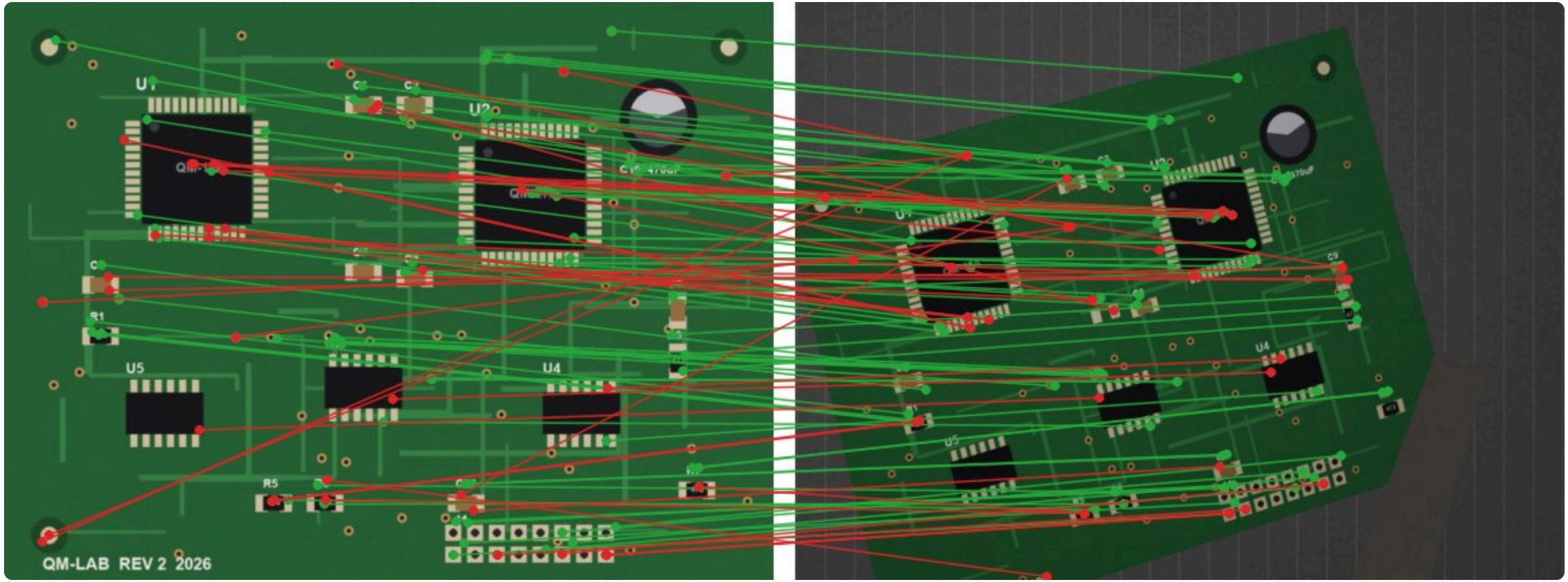
1 · Ratio test (Lowe)
Accept a pair only if the best match is clearly better than the second best: $d_1 < 0.75 \cdot d_2$ (Lowe proposed 0.8; 0.75 is a common, stricter choice).
A point that looks like several others — one of many identical chip pins, washers or screw heads — fails the test. Correctly: it cannot be told apart.

2 · RANSAC (Fischler and Bolles, 1981)
Pick 4 pairs at random and compute the transformation they imply (for a flat part: a homography). Count how many other pairs agree within 3 pixels. Repeat many times; keep the transformation with the most agreement. Pairs that disagree are **outliers**.

Test board → production image	pairs
Nearest neighbours	2,412
Pass the ratio test	905
Agree with the RANSAC transformation (inliers)	655
Rejected by RANSAC (outliers)	250

Matching the board on the conveyor

Rotated, smaller, darker, unevenly lit, partly covered by a gripper

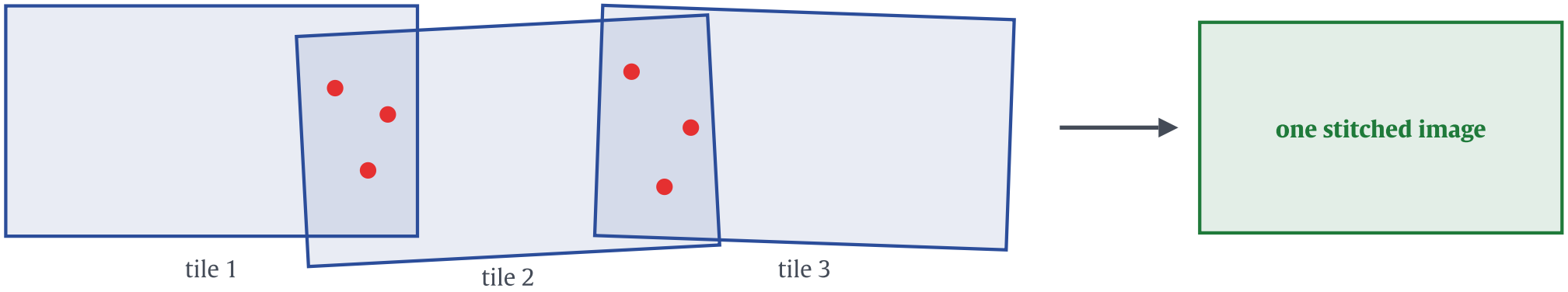


inliers: agree with the transformation (70 of 655 drawn) outliers: rejected (40 of 250 drawn)

The transformation estimated from the inliers is within **0.17 pixels** of the true one, everywhere on the board — although part of it is hidden and the light is different. That is the robustness to scale, rotation, light and clutter that template matching lacked.

Stitching: many images, one picture

Match interest points in the overlap, estimate the transformation, warp and blend



EVERYDAY · THE PANORAMA ON YOUR PHONE

As you sweep the phone, it matches interest points between consecutive frames, estimates how each frame moved, and warps them into one wide image. The seams you sometimes see are wrong or missing matches.

INDUSTRY · PARTS LARGER THAN THE FIELD OF VIEW

A car door panel, an aircraft skin panel, a whole PCB panel or a wind-turbine blade seen by a drone: a camera close enough to see small defects sees only part of it. The tiles are stitched into one image, and defects are located on the whole part.

Recap TAKEAWAYS

Part 4 — signature and matching

1. The SIFT descriptor: 4 × 4 cells × 8 gradient directions = 128 numbers, normalised — invariant to scale, rotation and brightness.
2. Descriptors are compared by a distance (SIFT: L2); each point gets its nearest neighbour in the other image.
3. The ratio test rejects points that look like several others; RANSAC rejects pairs that disagree with the geometry.
4. On the board on the conveyor: 655 correct correspondences, alignment within 0.17 pixels.
5. Stitching chains the same steps across overlapping images.

DISCUSS

You match a reference board to a production board: 40 pairs pass the ratio test, and RANSAC rejects 35 of them. What would you check first?

PART 5

Interest points in quality control

Where they earn their place on a production line

Five jobs on a production line

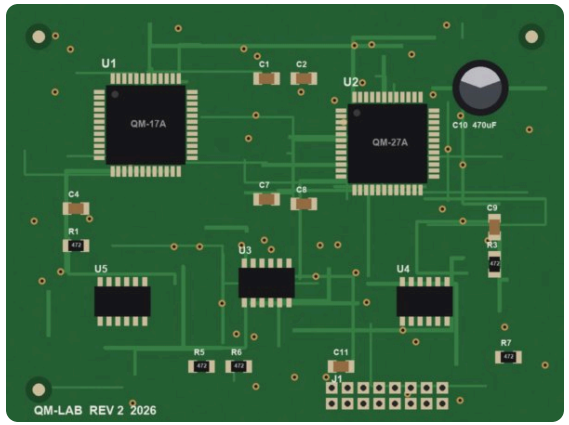
Where interest points do the work

Job	What the interest points do	Examples
Identify and sort	recognise which part, or which variant, is in view	mixed parts on one conveyor; the right label on the right box
Guide a robot	give position and orientation of the part for picking	bin picking, pick-and-place, screwing
Align before comparing	register the image onto a golden reference, so that differences mean defects	circuit boards, printed labels and packaging, castings
Stitch	build one image of a part larger than the field of view	body panels, aircraft skins, PCB panels, turbine blades
Measure in 3D, track motion	find the same point in two cameras, or at two instants	stereo and photogrammetry (the 3D shape measurement lecture); displacement and vibration from video (the course internship: Lucas–Kanade tracking with pyIDI)

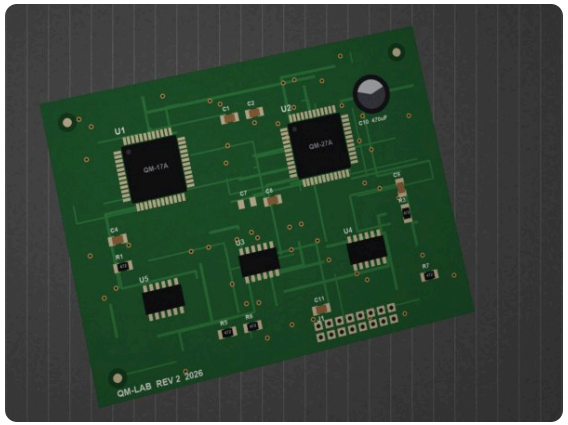
In every job the points are a **means**, not the result: the result is a transformation, a position, a stitched image or a displacement. Its accuracy depends on how many good correspondences there are and how well they are spread over the part.

Align, then compare

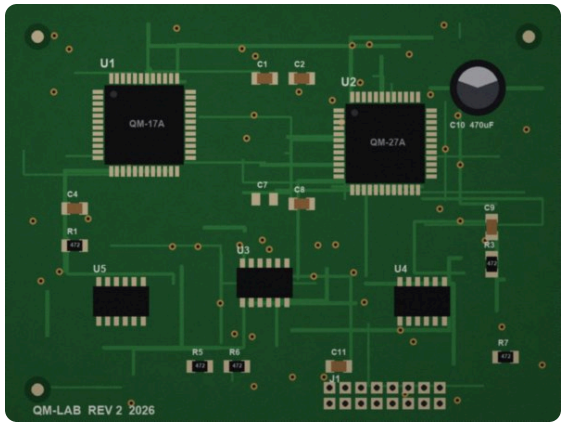
Finding a missing capacitor on a board that arrives at any angle



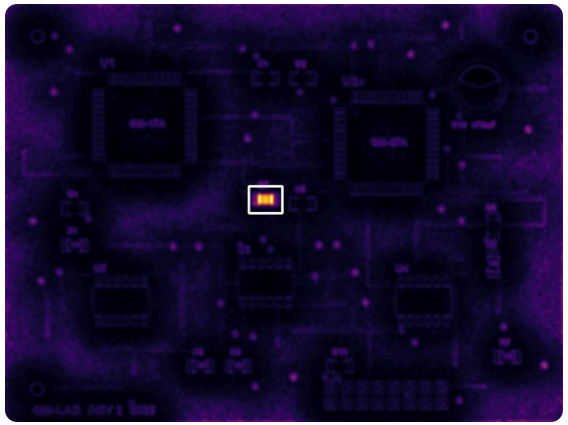
1 · golden reference



2 · production image



3 · warped onto the reference



4 · difference

1. Match interest points between reference and production image: **631** inliers.
2. Estimate the transformation with RANSAC: within **0.20 pixels** of the truth over the whole board.
3. Warp the production image onto the reference, normalise both for the local brightness, subtract.
4. One region stands out, about 4 × the background level: **C7, a missing capacitor**.

Automated optical inspection (AOI) of circuit boards works on this principle — with the fiducials as guaranteed interest points.

Interest points are measurements too

How well a detector finds the same point again — and when it cannot

- **Repeatability** — the share of points found again after a change of view or light. It is the detector's equivalent of a gauge's repeatability, and the standard way detectors are compared.
- **Localisation** — how precisely each point is placed. Refined below a pixel; with hundreds of points, the alignment on our board was within 0.20 pixels.
- **Distinctiveness** — how rarely two different points look alike: what the ratio test checks.

When natural points fail

Uniform or shiny surfaces — no texture, no points. **Repetitive patterns** — many identical candidates. **Strong perspective or deformation** — beyond what the model allows. **Motion blur** — the structure is smeared.

Then: design the points

Fiducials on circuit boards; coded targets in photogrammetry; a random **speckle pattern** sprayed on a part for digital image correlation, which measures strain from the motion of the speckles.

The family of detectors

Choose by invariance, speed and licence

Method	Finds	Invariant to	Speed	Licence
Harris (1988)	corners	rotation; not scale	fast	free
SIFT (Lowe, 2004)	blobs (difference of Gaussians); 128-number descriptor	scale, rotation, brightness	moderate	free since the patent expired in 2020
SURF (Bay et al., 2006)	blobs, with box filters	scale, rotation	faster than SIFT	patented
ORB (Rublee et al., 2011)	corners, with a 256-bit binary descriptor	rotation; scale through an image pyramid	very fast, real time on small hardware	free
Learned features (e.g. SuperPoint, 2018)	points and descriptors trained by a neural network	strong changes of view and light	needs a GPU	check: often for research use only

On a production line the choice is an engineering trade-off like any other: the cycle time fixes the speed, the variation of the parts fixes the invariance, and the licence decides whether the method can be shipped in a product.

Recap TAKEAWAYS

Part 5 — and the whole lecture in one line: find, describe, match, verify

1. Interest points let a machine identify, locate, align, stitch and measure parts that never sit in the same place twice.
2. Align, then compare: a board at any angle, registered within a fraction of a pixel, reveals a missing component by subtraction.
3. Interest points are measurements: judge a detector by its repeatability, localisation and distinctiveness.
4. Where nature gives no points, design them: fiducials, coded targets, speckle patterns.

Coming up
3D optical shape measurement: stereo cameras and photogrammetry need exactly these correspondences — the same point seen from two places gives its depth.

Try it
The figures of this lecture were made with OpenCV in Python: `cv2.SIFT_create()`, `knnMatch`, `findHomography(…, cv2.RANSAC)`. Three function calls.

Questions?

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github.com/LolloCappo/student-hub